# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 7 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 5: **Parallel and Intersecting Lines** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** A transversal makes co-interior angles in the ratio $1:5$ on parallel lines. The smaller angle is: 1

**(a)** $40^\circ$ **(b)** $20^\circ$ **(c)** $50^\circ$ **(d)** $30^\circ$

**2.** If $(x+15)^\circ$ and $(2x-6)^\circ$ are supplementary, $x$ equals: 1

**(a)** 67 **(b)** 57 **(c)** 52 **(d)** 62

**3.** If $l \parallel m$ and two co-interior angles are $123^\circ$ and $(x)^\circ$, then $x =$ 1

**(a)** $57^\circ$ **(b)** $67^\circ$ **(c)** $47^\circ$ **(d)** $77^\circ$

**4.** The angle which is 1 times its complement is: 1

**(a)** $65^\circ$ **(b)** $55^\circ$ **(c)** $45^\circ$ **(d)** $35^\circ$

**5.** Half of the supplement of $120^\circ$ is: 1

**(a)** $20^\circ$ **(b)** $30^\circ$ **(c)** $50^\circ$ **(d)** $40^\circ$

**6.** The angle bisector of a $150^\circ$ angle makes each part: 1

**(a)** $85^\circ$ **(b)** $95^\circ$ **(c)** $65^\circ$ **(d)** $75^\circ$

**7.** Three distinct lines pass through one point. The number of pairs of vertically opposite angles is: 1

**(a)** 3 **(b)** 6 **(c)** 4 **(d)** 2

**8.** A transversal makes co-interior angles in the ratio $4:5$ on parallel lines. The smaller angle is: 1

**(a)** $100^\circ$ **(b)** $90^\circ$ **(c)** $70^\circ$ **(d)** $80^\circ$

**9.** The angle which is 9 times its complement is: 1

**(a)** $81^\circ$ **(b)** $91^\circ$ **(c)** $71^\circ$ **(d)** $101^\circ$

**10.** The angle which is 17 times its complement is: 1

**(a)** $105^\circ$ **(b)** $75^\circ$ **(c)** $95^\circ$ **(d)** $85^\circ$

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Is $l \parallel m$ if a transversal gives corresponding angles $62^\circ$ and $62^\circ$? Give reason. 2

**12.** Find the angle that is $100^\circ$ less than its supplement. 2

**13.** Find $x$ if $(3x-8)^\circ$ and $(x+30)^\circ$ are vertically opposite angles. 2

**14.** Is $l \parallel m$ if a transversal gives corresponding angles $48^\circ$ and $84^\circ$? Give reason. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** A transversal cuts parallel lines. Interior angles on the same side are $(2x + 5)^\circ$ and $(3x + 25)^\circ$. Find $x$ and the two angles. **3**

**16.** In a pair of intersecting lines, the angles are $(x + 9)^\circ$, $(2x - 6)^\circ$ forming a linear pair. Find $x$ and the vertically opposite angles. **3**

**17.** An angle is $12^\circ$ more than twice its complement. Find the angle and its complement. **3**

**OR**

A transversal cuts parallel lines. Interior angles on the same side are $(2x + 10)^\circ$ and $(3x + 30)^\circ$. Find $x$ and the two angles. **3**

**18.** In a pair of intersecting lines, the angles are $(x + 21)^\circ$, $(2x - 3)^\circ$ forming a linear pair. Find $x$ and the vertically opposite angles. **3**

**OR**

An angle is $24^\circ$ more than twice its complement. Find the angle and its complement. **3**

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** Lines $l \parallel m$, transversal $t$. $\angle 1 = (2x + 8)^\circ$, $\angle 2 = (3x - 22)^\circ$ are alternate interior angles; $\angle 3$ is the co-interior angle of $\angle 1$. Find $x$, $\angle 1$, $\angle 2$ and $\angle 3$. **5**

**OR**

Around a point, angles are $x^\circ$, $2x^\circ$, $3x^\circ$ and $150^\circ$. Find $x$ and each angle, and say which are acute, right or obtuse. **5**

**20.** Lines $l \parallel m$, transversal $t$. $\angle 1 = (2x + 12)^\circ$, $\angle 2 = (3x - 18)^\circ$ are alternate interior angles; $\angle 3$ is the co-interior angle of $\angle 1$. Find $x$, $\angle 1$, $\angle 2$ and $\angle 3$. **5**

**OR**

Around a point, angles are $x^\circ$, $2x^\circ$, $3x^\circ$ and $36^\circ$. Find $x$ and each angle, and say which are acute, right or obtuse. **5**

## END OF QUESTION PAPER — Total: 40 Marks

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